

# Adam W. Duran, P.E., PMP

ASSISTANT TEACHING PROFESSOR

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[LinkedIn](#) · [ResearchGate](#) · [Google Scholar](#)

## EDUCATION

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|--------------------------------------------------------------------------------------------|------|
| <b>M.S., Engineering and Technology Management</b><br>Colorado School of Mines, Golden, CO | 2016 |
| <b>M.S., Engineering (Mechanical Specialty)</b><br>Colorado School of Mines, Golden, CO    | 2011 |
| <b>B.S., Engineering (Mechanical Specialty)</b><br>Colorado School of Mines, Golden, CO    | 2009 |

## LICENSURE & CERTIFICATIONS

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- **Professional Engineer (P.E.)**, State of Colorado, License No. PE.0061606 (2022)
- **Project Management Professional (PMP)**, Project Management Institute, ID 3726574 (2024)

## SCHOLARLY PROFILE

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- **Peer-reviewed publications and technical reports:** 30+
- **Google Scholar citations:** 1,000+   **h-index:** 17   **i10-index:** 20
- **Career competitive funding (PI / Proposal Lead):** \$10.5M+  
*Major components include: \$6.3M as Director, Shell GameChanger Accelerator (GCxN); \$2M as PI, Fleet DNA database; \$1M+ across DOE FOA vehicle technology awards at NREL; \$500k+ proposal portfolio at QS-2; \$520k+ DOE EcoCAR Innovation Challenge and \$255k total BWC program value at Mines.*
- **Courses taught at Mines:** 14 distinct courses
- **Courses developed or co-developed:** 8

Google Scholar: [scholar.google.com/citations?user=BQ404mAAAAAJ](https://scholar.google.com/citations?user=BQ404mAAAAAJ)

## ACADEMIC APPOINTMENTS

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**Assistant Teaching Professor**

Department of Mechanical Engineering, Colorado School of Mines

August 2022 – Present

Golden, CO

*Teaching and Program Leadership:*

- Develop and teach undergraduate and graduate courses in mechanical engineering, vehicle dynamics, aerospace systems engineering, combustion, and financial engineering; maintain 18+ credit annual teaching load (peak 21.0 FDR credits, CY23).
- Co-led expansion of the Mines Automotive Engineering Program to 252 track-declared undergraduates (17% of ME track-declared students), generating an estimated \$365K–\$827K in annual tuition<sup>1</sup>; program ranks 2nd within ME by enrollment, behind only Aerospace. Student-to-faculty

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<sup>1</sup>Estimated based on 252 students × 9–18 credits/year in track courses × prevailing in-state/out-of-state tuition rates. Actual revenue attribution depends on institutional accounting methodology.

ratio: 126:1 (with only two core faculty).

- Capstone Leadership Team, ME Co-Representative (Fall 2025 – Present): program-level planning, delivery standards, and coordination with the Industry Partnership Board Steering Committee.
- MEGN 300 Course Coordinator (Spring 2025): aligned multi-section instruction; mentored new instructors Kuska and Thomas.

#### *Service and Coordination:*

- FE Exam Review Coordinator (Spring 2023 – Present): organizes campuswide review each semester; recruits instructors; personally teaches Math, Prob/Stats, Ethics, and Materials Science sessions.
- Lead PI, DOE Battery Workforce Challenge (Spring 2024 – Present); Lead PI, DOE EcoCAR Innovation Challenge (Spring 2026 – Present); Co-PI: Prof. P. Brodsky. See Grants section.

### **Adjunct Instructor**

2016 – 2022

Colorado School of Mines (Multiple Departments)

Golden, CO

- Co-developed and taught MEGN 417 (Vehicle Dynamics) and MEGN 455 (Aerospace Systems Engineering) in the Department of Mechanical Engineering.
- Co-developed launch-year curriculum for the UHSP Grandey Honors Engineering program (HNRS 110/120 Leadership by Design); built SolidWorks CAD curriculum and facilitated 3+ workshops of ~100 students/session.
- Taught HNRS 105/115 IDEAS and EDS 151 Cornerstone Design; served as Project Advisor and Client for EDS 491/492 Senior Capstone Design teams.

## **PROFESSIONAL EXPERIENCE**

### **Senior Engineer**

2020 – 2022

Quantitative Scientific Solutions (QS-2)

Arlington, VA (Remote)

- Led research on automated driving systems (ADS) and advanced driver assistance systems (ADAS), covering enabling technologies, policy, and standards.
- Led proposal development process from scoping through work capture and execution; captured and led a portfolio in excess of \$500k cumulative funding.
- Spearheaded cultivation of partnerships and stakeholder engagement activities including workshops and conferences; managed project budgets and staff.

### **Program Manager, Innovation & Entrepreneurship Center**

2018 – 2020

National Renewable Energy Laboratory (NREL)

Golden, CO

- Cofounder and Director, Shell GameChanger Accelerator Powered by NREL (GCxN): directed a \$6.3M clean technology startup accelerator with full responsibility for program strategy, P&L, partner agreements, and participant mentorship.
- Portfolio outcomes (as of 2025 Annual Report): \$1.048B raised by cohort companies, 954 new startup hires (374% growth since 2022), average TRL progression from 4.1 to 5.9, \$140:1 leverage ratio on Shell project funding, 30 companies in portfolio.
- Managed contractual relationships with Shell International and NREL; assembled startup cohorts through rigorous down-selection; built and led a team of technical, project management, and communications professionals.

**Senior Research Engineer, Transportation Data Science***2014 – 2018**NREL Transportation & Hydrogen Systems Center*

Golden, CO

- Led technical development and deployment of NREL's **Fleet DNA** commercial vehicle database (\$2M, 5 years): a national-scale repository for medium- and heavy-duty vehicle data used by fleet managers, OEMs, and researchers nationwide.
- PI for multiple DOE-funded projects: Class 8 truck electrification pathways (Cummins TSA, \$250k); multi-modal freight energy optimization (ORNL, \$250k); multi-speed transmission development (Eaton; DOE FOA, \$250k); extended-range Class 6 truck platform (Cummins/PACCAR; DOE FOA, \$250k).
- Applied statistical analysis and machine learning to develop duty cycles incorporated into EPA's Phase II Greenhouse Gas regulations for commercial vehicles.

**Research Engineer***2009 – 2014**NREL Transportation & Hydrogen Systems Center*

Golden, CO

- Managed City of Indianapolis Refuse Truck Fleet Evaluation (Clean Cities program): field data collection, drive cycle analysis, and simulation across HEV, HHV, and CNG options.
- Developed large-scale data analysis capabilities and published quarterly reports for a national EV data collection project (ARRA funding).

**Student Research Assistant***2008 – 2009**NREL Transportation & Hydrogen Systems Center*

Golden, CO

- Developed NREL's DRIVE<sup>TM</sup> tool: GUI-based duty cycle analysis and generation tool; commercialized via licensing and applied to numerous national fleet evaluation projects.

**TEACHING****Courses Taught**<sup>†</sup> Indicates course developed or co-developed by the instructor.**As Assistant Teaching Professor, August 2022 – Present:**

- **MEGN 417/527<sup>†</sup>**: Vehicle Dynamics and Powertrain Systems (UG/G) (F22, F23, F24, F25)
- **MEGN 455<sup>†</sup>**: Aerospace Systems Engineering (UG) (F22, F23, F24)
- **MEGN 456<sup>†</sup>**: Space Operations and Mission Design (UG) (S23, S24, S25)
- **MEGN 566<sup>†</sup>**: Combustion, Online Graduate Elective (G) (F23, F24)
- **MEGN 301<sup>†</sup>**: Mechanical Integration and Design (UG) (SSI25, S26, SSI26)
- **MEGN 300**: Instrumentation and Automation (UG) (S23, S24; Course Coordinator, S25)
- **MEGN 200**: Introduction to Mechanical Engineering: Programming and Hardware Interface (UG) (F25)
- **EBGN 599A<sup>†</sup>**: Financial Engineering for Engineers (G) (S26)
- **EDS 491/492**: Senior Capstone Design (UG) (F22, F23, F24, F25; S23, S24, S25, S26)
- **MEGN 598/599**: Independent Study (UG/G) (multiple students; see Student Mentoring)

**As Adjunct Instructor, August 2016 – August 2022:**

- **MEGN 417<sup>†</sup>**: Vehicle Dynamics and Powertrain Systems (co-developed with Prof. Bogin)
- **MEGN 455<sup>†</sup>**: Aerospace Systems Engineering (co-developed with Prof. Jennifer Erickson)
- **HNRS 105/115**: IDEAS I & II (co-taught)
- **HNRS 110/120<sup>†</sup>**: Leadership by Design I & II (co-developed, launch year)
- **EDS 491/492**: Senior Capstone Design (Project Advisor)
- **EDS 151**: Cornerstone Design

### Course Development and Pedagogical Innovations

- **Experiential Mastery-Based Learning (EMBL) Framework** (MEGN 417/527): specifications-based competency system with six Course Learning Standards, four mastery levels, bundle-based grade determination, and a structured revision token system. Grounded in Bloom (1968), Nilson (2015), Wiggins & McTighe (2005), and Kolb (1984); refined through Trefny Engineering Learning Fellows peer review.
- **Scenario-Based Industry Project Formats**: MEGN 456 uses a semester-long NASA RFP mission design project (orbital mechanics, sensor trade studies, full SE documentation package); MEGN 455 uses a live SBIR Phase I white paper format, mirroring federal proposal practice.
- **EBGN 599A Financial Engineering** (new course, S26): 16-week curriculum developing professional quantitative finance skills in Python, including walk-forward validation, Mahalanobis distance regime detection, and tail hedging.
- **Zero Textbook Cost (ZTC)**: all required texts for MEGN 417, MEGN 566, and MEGN 301 available at no cost through Mines Library; MEGN 456 supplementary readings ZTC-designated.
- **MEGN 566 Online Redesign**: iterative redesign across two offerings (F23, F24) testing hypotheses about asynchronous self-regulation for working professionals; introduced synchronous check-ins, worked examples, and revised pacing.

### Teaching-Related Professional Development

**2025–26**: Trefny Center Engineering Learning Fellow; KEEN Crescendo Program (1 of 5 national members); KERN Community Catalysts, Cohort 13 (EMxAI Working Group Leader); Human-Centered AI Intensive (Trefny); KEEN National Conference; KERN/KEEN EMxAI Summit (Presenter); KEEN EML Mini-Workshop (produced activity card); KERN Making with AI Workshop (produced activity card); Teaching with TAs Workshop (Trefny).

**2024**: Alternative Grading Intensive (Trefny); Princeton Combustion Summer School; Discipline-Based Education Research (DBER) Workshop Series.

**2023–24**: Trefny New Faculty Learning Community; Foundations of Course Design Follow-Up Training; Peer Observation with Teaching Prof. Chris Shorey; ASIST Suicide Intervention Workshop.

**2022**: Trefny EFOL Training (MEGN 566 Online); Trefny ELFCD Training (MEGN 456); Faculty Standards Review Committee Training (Mines Online); COPUS Classroom Observation (confirmed student-centered practice); National Collegiate Honors Conference (NCHC) Participant (2021).

### GRANTS & EXTERNAL FUNDING

#### Active Federal and Industry Grants (Mines Appointment)

- **DOE EcoCAR Innovation Challenge** (awarded Spring 2026, multi-year, ≈\$520,000)  
Lead PI: Duran; Co-PI: Prof. P. Brodsky  
One of DOE's flagship Advanced Vehicle Technology Competition programs; Mines selected as one of a limited number of participating institutions nationwide.

- **DOE / Stellantis Battery Workforce Challenge** (2023 – Present)

DOE/Stellantis base grant: \$147,000 (3 years); 50% PI allocation; Co-PI: Prof. P. Brodsky

Chevron industry sponsorship: \$20,000

Student travel stipends (through Mines): \$35,600

Competition prizes (awarded to students): \$32,500

**Total BWC Program Value: \$255,100**

Additional significant in-kind value from software (GT-Suite, Siemens), hardware (Analog Devices, Dana), and vehicle donation (Dodge RAM ProMaster Van) not included.

Results: 1st place nationally Year 1 (2023–24); 4th place nationally Year 2 (2024–25); more than a dozen technical and project management awards.

### Pedagogical and Internal Grants (Mines Appointment)

- **Mines Technology Fee:** Shock Dynamometer & Spring Rater (Spring 2023): \$11,900; with Prof. P. Brodsky
- **Mines Technology Fee:** RC Cars & Batteries (Spring 2025): \$5,900; with Prof. P. Brodsky
- **Mines Technology Fee:** RC Dyno & Battery Cycler (Fall 2025): \$13,800; with Prof. P. Brodsky
- **KEEN EML Mini-Grant** (Spring 2025): \$600; produced two activity cards disseminated to the national KEEN community
- **Mines Library Zero Textbook Cost Initiative:** MEGN 417, MEGN 456, MEGN 566, MEGN 301 (2022 – Present)

### Prior Career Funding (Pre-Mines, NREL and QS-2)

- Shell GameChanger Accelerator (GCxN), Director: \$6.3M (5 years)
- Fleet DNA Commercial Vehicle Database, PI: \$2M (5 years)
- Exploring Pathways to Class 8 Truck Electrification (Cummins TSA), PI: \$250k
- Multi-Modal Freight Energy Optimization (ORNL collaboration), PI: \$250k
- Development of Multi-Speed Transmission for PDEV (Eaton; DOE FOA), NREL PI: \$250k
- Development of Extended-Range ETREE Truck (Cummins/PACCAR; DOE FOA), NREL PI: \$250k
- Automated Driving Systems Research Portfolio (QS-2), Proposal Lead: \$500k+
- **Total career competitive funding (PI / Proposal Lead): \$10.5M+**

## PUBLICATIONS

*Author's name in **bold**. Listed in reverse chronological order within each category.*

### Journal Articles (Peer-Reviewed)

1. Mistry, A.N., Brodsky, P., Brickey, A.J., **Duran, A.**, Mozur, E., Rajasegar, R., Jackson, G., DeCaluwe, S.C., Kee, R.J., O'Hayre, R., Newman, A., Braun, R., Bazilian, M., & Sullivan, N.P. "Not-So-Quick and Not-So-Dirty Solutions to Decarbonize Off-Road Vehicles." *ACS Energy Letters*, 2025, 10(11), 5322–5327. DOI: [10.1021/acsenenergylett.5c02459](https://doi.org/10.1021/acsenenergylett.5c02459)
2. Mistry, A.N., Brodsky, P., **Duran, A.**, Mozur, E., Rajasegar, R., Brickey, A.J., Newman, A., Ingram, B.J., Braun, R., O'Hayre, R., Jackson, G., DeCaluwe, S.C., Kee, R.J., Bazilian, M., & Sullivan, N.P. "Electrifying Off-Road Vehicles: Is 1000 Wh/kg Enough?" *ACS Energy Letters*, 2024, 9(8), 4053–4058. DOI: [10.1021/acsenenergylett.4c01276](https://doi.org/10.1021/acsenenergylett.4c01276)
3. Miller, E. and **Duran, A.** "A Deterministic Multivariate Clustering Method for Drive Cycle Generation from In-Use Vehicle Data." *SAE Technical Paper* 2021-01-0395, 2021. DOI: [10.4271/2021-01-0395](https://doi.org/10.4271/2021-01-0395)
4. **Duran, A.**, Li, K., Kresse, J., & Kelly, K. "Development of 80 and 100 Mile Work Day Cycles



- Representative of Commercial Pickup and Delivery Operation.” *SAE Technical Paper* 2018-01-1192, 2018. DOI: [10.4271/2018-01-1192](https://doi.org/10.4271/2018-01-1192)
5. **Duran, A.**, Phillips, C., Perr-Sauer, J., Kelly, K., & Konan, A. “Leveraging Big Data Analysis Techniques for US Vocational Vehicle Drive Cycle Characterization, Segmentation, and Development.” *SAE Technical Paper* 2018-01-1199, 2018. DOI: [10.4271/2018-01-1199](https://doi.org/10.4271/2018-01-1199) (*Drive cycles adopted by U.S. EPA in GHG Phase 2 rulemaking.*)
  6. Lammert, M., Bugbee, B., Hou, Y., Mack, A., Muratori, M., Holden, J., **Duran, A.**, & Swaney, E. “Analysis of Opportunities for Truck Platooning from Telematics Data.” *SAE Technical Paper* 2018-01-1083, 2018. DOI: [10.4271/2018-01-1083](https://doi.org/10.4271/2018-01-1083)
  7. Muratori, M., Holden, J., Lammert, M., **Duran, A.**, et al. “Potentials for Platooning in U.S. Highway Freight Transport.” *SAE International Journal of Commercial Vehicles*, 10(1), 2017. DOI: [10.4271/2017-01-0086](https://doi.org/10.4271/2017-01-0086)
  8. Miller, E., Konan, A., & **Duran, A.** “Bayesian Parameter Estimation for Heavy-Duty Vehicles.” *SAE Technical Paper* 2017-01-0528, 2017. DOI: [10.4271/2017-01-0528](https://doi.org/10.4271/2017-01-0528)
  9. Wang, L., **Duran, A.**, Kelly, K., Konan, A., et al. “The Evaluation of the Impact of New Technologies for Different Powertrain Medium-Duty Trucks on Fuel Consumption.” *SAE Technical Paper* 2016-01-8134, 2016. DOI: [10.4271/2016-01-8134](https://doi.org/10.4271/2016-01-8134)
  10. Wood, E., **Duran, A.**, & Kelly, K. “EPA GHG Certification of Medium- and Heavy-Duty Vehicles: Development of Road Grade Profiles Representative of US Controlled Access Highways.” *SAE International Journal of Commercial Vehicles*, 9(2):94–104, 2016. DOI: [10.4271/2016-01-8017](https://doi.org/10.4271/2016-01-8017)
  11. Marcinkoski, J., Vijayagopal, R., Kast, J., & **Duran, A.** “Driving an Industry: Medium and Heavy Duty Fuel Cell Electric Truck Component Sizing.” *World Electric Vehicle Journal*, 8(1), 2016.
  12. Wang, L., **Duran, A.**, Gonder, J., & Kelly, K. “Modeling Heavy/Medium-Duty Fuel Consumption Based on Drive Cycle Properties.” *SAE Technical Paper* 2015-01-2812, 2015. DOI: [10.4271/2015-01-2812](https://doi.org/10.4271/2015-01-2812)
  13. Wang, L., Kelly, K., Walkowicz, K., & **Duran, A.** “Quantitative Effects of Vehicle Parameters on Fuel Consumption for Heavy-Duty Vehicle.” *SAE Technical Paper* 2015-01-2773, 2015. DOI: [10.4271/2015-01-2773](https://doi.org/10.4271/2015-01-2773)
  14. Lopp, S., Wood, E., & **Duran, A.** “Evaluating the Impact of Road Grade on Simulated Commercial Vehicle Fuel Economy Using Real-World Drive Cycles.” *SAE Technical Paper* 2015-01-2739, 2015. DOI: [10.4271/2015-01-2739](https://doi.org/10.4271/2015-01-2739)
  15. Lammert, M., Burton, J., Sindler, P., & **Duran, A.** “Hydraulic Hybrid and Conventional Parcel Delivery Vehicles’ Measured Laboratory Fuel Economy on Targeted Drive Cycles.” *SAE International Journal of Alternative Powertrains*, 4(1), 2015. DOI: [10.4271/2014-01-2375](https://doi.org/10.4271/2014-01-2375)
  16. **Duran, A.**, Ragatz, A., Prohaska, R., Kelly, K., & Walkowicz, K. “Characterization of In-Use Medium Duty Electric Vehicle Driving and Charging Behavior.” *IEEE International Electric Vehicle Conference (IEVC)*, 2014. DOI: [10.1109/IEVC.2014.7056213](https://doi.org/10.1109/IEVC.2014.7056213)
  17. Lammert, M., **Duran, A.**, Diez, J., Burton, K., et al. “Effect of Platooning on Fuel Consumption of Class 8 Vehicles Over a Range of Speeds, Following Distances, and Mass.” *SAE International Journal of Commercial Vehicles*, 7(2):626–639, 2014. DOI: [10.4271/2014-01-2438](https://doi.org/10.4271/2014-01-2438)
  18. Wood, E., Burton, E., **Duran, A.**, & Gonder, J. “Contribution of Road Grade to the Energy Use of Modern Automobiles Across Large Datasets of Real-World Drive Cycles.” *SAE Technical Paper* 2014-01-1789, 2014. DOI: [10.4271/2014-01-1789](https://doi.org/10.4271/2014-01-1789)
  19. Burton, J., Walkowicz, K., Sindler, P., & **Duran, A.** “In-Use and Vehicle Dynamometer Evaluation and Comparison of Class 7 Hybrid Electric and Conventional Diesel Delivery Trucks.” *SAE International Journal of Commercial Vehicles*, 6(2), 2013. DOI: [10.4271/2013-01-2468](https://doi.org/10.4271/2013-01-2468)

20. **Duran, A.** & Walkowicz, K. “A Statistical Characterization of School Bus Drive Cycles Collected via Onboard Logging Systems.” *SAE International Journal of Commercial Vehicles*, 6(2), 2013. DOI: [10.4271/2013-01-2400](https://doi.org/10.4271/2013-01-2400)
21. Lammert, M., Walkowicz, K., **Duran, A.**, & Sindler, P. “Measured Laboratory and In-Use Fuel Economy Observed over Targeted Drive Cycles for Comparable Hybrid and Conventional Package Delivery Vehicles.” *SAE Technical Paper* 2012-01-2049, 2012. DOI: [10.4271/2012-01-2049](https://doi.org/10.4271/2012-01-2049)
22. **Duran, A.** & Earleywine, M. “GPS Data Filtration Method for Drive Cycle Analysis Applications.” *SAE Technical Paper* 2012-01-0743, 2012. DOI: [10.4271/2012-01-0743](https://doi.org/10.4271/2012-01-0743)

### Book Chapters

1. **Duran, A.** “Hydrogen Dreams, False Claims: The Ethical Collapse of Nikola Corporation.” Book chapter case study, Elgar Publishing, 2026.

### Conference Papers and Proceedings

1. Hou, Y., Moore, A.M., **Duran, A.W.**, Walkowicz, K.A., & Smith, D. “A Hybrid Tour-Based Model for Energy Analysis of Multi-Modal Intra-City Freight: A Case Study of Autonomous Electric Vehicles.” *98th Annual Meeting of the Transportation Research Board (TRB)*, Washington, D.C., January 2019.
2. Prohaska, R., **Duran, A.**, & Ragatz, A. “Statistical Characterization of Medium-Duty Electric Vehicle Drive Cycles.” *EVS28: 28th International Electric Vehicle Symposium and Exhibition*, May 2015.

### Technical Reports

1. Kimmel, S., **Duran, A.**, Robertson, J., Vanderveen, M., & Wendling, B. (2021). *Physical and Digital Infrastructure for Connected and Automated Vehicles (CAV): Code Framework*. Canadian Standards Association, Toronto, ON.
2. Perr-Sauer, J., **Duran, A.**, & Phillips, C. (2020). *Clustering Analysis of Commercial Vehicles Using Automatically Extracted Features from Time Series Data*. NREL/TP-2C00-74212. [nrel.gov/docs/fy20osti/74212.pdf](https://www.nrel.gov/docs/fy20osti/74212.pdf)
3. **Duran, A.**, Phillips, C., Konan, A., & Kelly, K. (2017). *The Development of Vocational Vehicle Drive Cycles and Segmentation*. NREL/TP-5400-65921. [nrel.gov/docs/fy17osti/65921.pdf](https://www.nrel.gov/docs/fy17osti/65921.pdf)
4. Konan, A., **Duran, A.**, Kelly, K., Miller, E., & Prohaska, R. (2017). *Characterization of PTO and Idle Behavior for Utility Vehicles*. NREL/TP-5400-66747. [nrel.gov/docs/fy17osti/66747.pdf](https://www.nrel.gov/docs/fy17osti/66747.pdf)
5. Wood, E., **Duran, A.**, Burton, E., Gonder, J., & Kelly, K. (2015). *EPA GHG Certification of Medium- and Heavy-Duty Vehicles: Development of Road Grade Profiles Representative of US Controlled Access Highways*. NREL Report No. TP-5400-63853.
6. Thornton, M., **Duran, A.**, Ragatz, A., Cosgrove, J., Sindler, P., Russel, R., & Johnson, K. (2015). *Data Collection, Testing, and Analysis of Hybrid Electric Trucks and Buses Operating in California Fleets*. NREL Report No. TP-5400-62009.
7. Wood, E., Burton, E., **Duran, A.**, & Gonder, J. (2014). *Appending High-Resolution Elevation Data to GPS Speed Traces for Vehicle Energy Modeling and Simulation*. NREL Report No. TP-5400-61109.
8. **Duran, A.**, Burton, E., Kelly, K., & Walkowicz, K. (2014). *Fleet DNA Project: Data Dictionary for Public Download Files*. NREL/TP-5400-62572. [nrel.gov/docs/fy14osti/62572.pdf](https://www.nrel.gov/docs/fy14osti/62572.pdf)

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## SOFTWARE, TOOLS, & INTELLECTUAL PROPERTY

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1. **NREL DRIVE™ Tool**: Copyright: “Duty Cycle Software Model,” 2008. Alliance for Sustainable Energy, LLC. TXu001725619. NREL Record of Invention CR/09-61.
2. **NREL Fleet DNA Database**: NREL Record of Invention SWR-13-00027, 2013.
3. **NREL Road Grade Analysis Algorithms**: NREL Record of Invention SWR-13-00032, 2013.
4. **Vehicle Fuel Economy Estimation Tool**: NREL Record of Invention SWR-14-21.
5. **Parameter Estimator for Engineering Systems**: NREL Record of Invention SWR-16-38.
6. **Clustering Analysis Code Artifact**: NREL Record of Invention SWR-21-96.
7. **Non-Alcoholic Spirit Concentrate**: Patent pending (Method & Muse Spirits, 2024–).

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## PRESENTATIONS & INVITED TALKS

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1. **Duran, A.** “Entrepreneurial Mindset in Engineering Education.” KEEN Crescendo Program, University of the Pacific, March 2026.
2. **Duran, A.** “Co-Creating with AI: New Frontiers in Teaching and Design.” Flash Talk, *Generative AI and Human Futures: A Summit on Teaching and Learning at Mines*, Colorado School of Mines, Fall 2025.
3. 15+ KEEN Entrepreneurial Mindset Learning community activity cards published (2025). Disseminated to national KEEN network.
4. KERN/KEEN EMxAI Summit, Presenter (February 2025).
5. SAE World Congress 2018: Leveraging Big Data Analysis Techniques for US Vocational Vehicle Drive Cycle Characterization, Segmentation, and Development.
6. SAE World Congress 2018: Development of 80 and 100 Mile Work Day Cycles Representative of Commercial Pickup and Delivery Operation.
7. 28th Coordinating Research Council Workshop 2018: Leveraging Big Data Analysis Techniques for US Vocational Vehicle Drive Cycle Characterization.
8. DOE Annual Merit Review 2017: Opportunities for Improving the Energy Efficiency of Multi-Modal Intra-City Freight Movement.
9. SAE Commercial Vehicle Congress 2016: EPA GHG Certification of Medium- and Heavy-Duty Vehicles: Development of Road Grade Profiles.
10. Beyond Light Duty Vehicle Electrification of Goods and People Movement: Workshop 2016.
11. DOE Annual Merit Review 2015: Fleet DNA Phase 1 Refinement & Phase 2 Implementation.
12. SAE Commercial Vehicles Congress 2015: Evaluating the Impact of Grade on Simulated Commercial Vehicle Fuel Economy.
13. IEEE International Electric Vehicle Conference 2014: Characterization of In-Use Medium Duty Electric Vehicle Driving and Charging Behavior.
14. DOE Annual Merit Review 2014: Poster Session, Fleet DNA.
15. 2014 NTEA Green Truck Summit: Data Driven Decision Making Tools.
16. DOE Annual Merit Review 2013: Poster Session, Fleet DNA.
17. SAE Commercial Vehicles Congress 2012: Vehicle Duty Cycles and Their Role in Design and Evaluation of Advanced Vehicle Technologies.
18. SAE World Congress 2012: GPS Data Filtration Method for Drive Cycle Analysis Applications.
19. Hybrid Truck Users Forum 2012: The What, Why, and How of Drive Cycles.
20. Automotive UI 2012, Tutorial: Introduction to Onboard Vehicular GPS Data Collection, Analysis, and Reporting.



## STUDENT MENTORING

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### Graduate Thesis and Dissertation Committees

- M.S. Thesis Co-Advisor: Aidan Strang, “Dynamics and Robotic Stabilization of a Centrifugal Hydroponic System” (Spring 2026 – Present; active)
- M.S. Thesis Committee Member: Daniel Azbell; successfully defended Spring 2026
- Ph.D. Committee Member: Kayden Tucker (Spring 2025 – Present; active)
- Ph.D. Committee Member: Finnegan Wilson (Spring 2024 – Present; active)

### Independent Study Students (MEGN 598/599)

*18 students supervised since August 2022; grouped by research area.*

#### Vehicle Dynamics and Automotive Systems (7 students):

- Quin Guy: Chassis Dynamometer Redesign for EV RC Cars (S23)
- Jacob Chwialkowski: Exploration in Damping Testing (Su23)
- Isaac Wheelis: Two-Wheel Vehicle Dynamics (S25)
- Kyle Coleman: Engine Laboratory Design & Management (S25)
- Jack Loveland: Exploration of Trailer Dynamics (S25)
- Louis Bartels: Shock Dynamometer Research, MEGN 598 (S25)
- Ben Stern: Shock Dynamometer Research, MEGN 598 (S25)

#### Battery Workforce Challenge / Competition Program Management (4 students):

- Colten Kristie: BWC Program Management (S24)
- Noam Ayalon: BWC Project Management; earned CAPM certification (F24)
- Wesley Swieter: Competition Project Management (S25); earned PMP certification
- Viviana Nelson: Competition Project Management (S26)

#### Space Systems and Mission Design (1 student):

- Sammy Shoun: Space Operations & Mission Design (Su23)

#### Current Graduate Research and Independent Study (F25–S26):

- Aidan Strang: MS Thesis co-advisor, “Dynamics and Robotic Stabilization of a Centrifugal Hydroponic System” (F25–present)
- Aaron Puffenberger: Vehicle Dynamics Research, MEGN 599 (F25, S26)
- Christopher Dahlgren: Vehicle Systems Research, MEGN 599 (F25, S26)
- Leah Schartz: Shock Dynamometer Research, MEGN 599 (F25)
- Eric Bueno Corral: Suspension Design and Modeling, MEGN 599 (S26)
- Connor Wilde: Financial Engineering Independent Study, EBGN 599 (S26)

#### Student Group Advising

- **Battery Workforce Challenge (BWC)** (Summer 2023 – Present), Lead PI (Spring 2024 – Present). Co-advisor: Prof. P. Brodsky. 1st place nationally, Year 1 (2023–24); 4th place nationally, Year 2 (2024–25); more than a dozen technical and project management awards across both years; \$80,100+ in student prizes and stipends; 2 MS and 3 BS students supported per program year.
- **Formula SAE** (Fall 2022 – Present), Co-advisor with Prof. P. Brodsky. 53rd of 120 international teams (2025 competition); completed all dynamic events including endurance race for the first time since 2016; team internship rate exceeding 90% (Summer 2024).

## Undergraduate Academic Advising

Advising load grew from 15 declared students (CY22) to 57 (CY24), tracking with growth of the Mines Automotive Engineering track. Write 5–10 recommendation letters per academic year.

## Senior Capstone Design Mentoring

Served as Client ( $\approx 22$  projects), Project Advisor ( $\approx 5$  projects), and Technical Advisor ( $\approx 6$  projects) across 30+ teams across 8 semesters (F22–S26). Notable project outcomes include 1st place (EV Powertrain Swap, F23) and 2nd place finishes (EV Powertrain Swap, F24; NASA Lunabotics, F23).

## SERVICE

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### Departmental Service (Mines ME)

#### *Leadership roles:*

- Capstone Leadership Team, ME Co-Representative (Fall 2025 – Present)
- Lead Faculty, ME Aerospace Elective Track (Fall 2025 – Present)
- FE Exam Review Coordinator and Lead Instructor (Spring 2023 – Present)

#### *Committee service:*

- ME Mechatronics Faculty Search Committee (Fall 2024)
- ME Solid Mechanics Faculty Search Committee (Spring 2026)
- ME Undergraduate Recruitment Committee (Fall 2024, Spring 2025)
- EDS Teaching Associate Professor Search Committee (Spring 2023)
- ME Online Programs Committee (Fall 2022)
- ME Wellness Committee (August 2022 – Present)
- Undergraduate Curriculum Committee, Stand-In Representative (F22–Sp23)

#### *Mentoring and curriculum support:*

- Instructor Mentoring: Jorge Ramirez (MEGN 300, Sp24); Allison Caster (MEGN 466, Sp24); Elijah Kuska and Siby Thomas (MEGN 300, Sp25); Jennifer Erickson (MEGN 455/456)
- UHSP SolidWorks Curriculum Development: 3+ workshops,  $\sim 100$  students/session (2021–2023)

### University Service (Mines)

- Capstone Industry Partnership Board Steering Committee (Fall 2025 – Present)
- GGE Promotion and Tenure Committee, External Member (Fall 2025)
- Presidential Committee for Faculty Success and Retention (Spring 2024 – Spring 2025)
- Faculty Handbook Review, IP and Copyright Sections 10.1/10.3 (Fall 2023)
- Faculty Senate Online Standards Committee (Fall 2022)
- Mines Online/OLED Course Reviewer: 12+ courses (2022–2026)
- Judge, Capstone Design Showcase (Fall 2022, Fall 2023)
- Faculty Connections Participant, CSM 101 (multiple semesters)

### Professional and Community Service

- SAE Handling Editor, Non-Event Technical Papers (Fall 2022 – Present)
- Reviewer: SAE, IEEE, TRB, ASEE, *Transportation Research Part D*, *Journal of Sustainable Transportation*, *Energy Environment & Policy*, *International Journal of Commercial Vehicles*
- BWC Faculty Advisory Board (Fall 2024)

- State of Colorado CEVC ZEV Workforce Development Group (2024)
- State of Colorado ZEV Mechanic Focus Team (2024)
- KERN Community Catalysts, Cohort 13 / EMxAI Working Group Leader (2025 –)
- KEEN Crescendo National Team (Spring 2026)
- Advisory Board Member, Orbital Mining Corporation
- AIAA Pathfinder Conference (September 2023)
- Guest Lecturer: MEGN 408 (F23), MEGN 452 (F25), HNRS 110 (F24), MEGN 391 (F24, F25)
- Guest Judge, Technology Entrepreneurship EBGN 566 (F24)
- Green Mountain High School Innovation Fair Judge (April 2023)
- Code Denver: High School Arduino Facilitator (Su23)
- Mines Moonlight Breakfast Volunteer (May 2023)

### Professional Memberships

ASEE · SAE · AIAA · IEEE · TRB · PMI

## HONORS & AWARDS

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- **2026:** Mines Profound: 3 student project awards advised
- **2026:** KEEN Crescendo Program: selected as 1 of 5 national members
- **2025:** Mines Trefny Center Engineering Learning Fellow
- **2025:** Kern Family Foundation Community Catalyst Program, Cohort 13
- **2025:** DOE Battery Workforce Challenge: 4th Place Nationally (\$8k+ prize money to team)
- **2025:** KEEN EML Mini-Grant: \$600
- **2024:** Project Management Professional (PMP) Certification
- **2024:** DOE Battery Workforce Challenge: 1st Place Nationally (\$10k+ prize money to team)
- **2023:** Mines Zero Textbook Cost Award (MEGN 456, MEGN 566)
- **2022:** Professional Engineer (P.E.), State of Colorado
- **2019:** NREL Outstanding Partnership Award (GCxN)
- **2014:** NREL President's Award: Outstanding efforts to enhance the Transportation Secure Data Center and Fleet DNA repositories

## ENTREPRENEURIAL VENTURES

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### Co-Founder

*Method & Muse Spirits*

*2023 – Present*

Golden, CO

- Award-winning craft distillery co-founded with Prof. Aubrey Wigner (Mines) and four Mines alumni; operates on Mines' 45th Drive campus.
- Patent-pending non-alcoholic spirit concentrate (filed 2024).
- Submitted institutional partnership proposal to Mines (March 2026) scoped across professional education programming and a credit-bearing distilling science course targeting Fall 2026 pilot enrollment.
- Activity conducted in accordance with Mines outside employment and conflict-of-interest policies.

## MEDIA ENGAGEMENT

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1. *CleanTechnica* (2020): "Shell Has a Bigger Clean Energy Plan Than You Think"
2. *Energy Storage News* (2020): "Shell-NREL Accelerator: Important to Explore Alternatives to Lithium-Ion"

3. *GreenTech Media* (2020): “From the Lab: The Energy Transition Needs Perovskite PV and Alternative Energy Storage”
4. *Innovation and Tech Today* (2020): “How Resilience Technology is Helping Communities Through Natural Disasters”
5. *Scientific American* (2019): “Climate Concerns Are Pushing Oil Majors to Look Beyond Fossil Fuels”
6. *Medium.com* (2019): “A Global Energy Gamechanger”
7. *Service Truck Magazine* (2019): “How to Make Smart Choices when Acquiring a New Truck”
8. *EurekAlert!* (2015): “NREL Collaborates with Trucking Industry to Prioritize R&D Opportunities”